



CONDA[®]

Advanced Molecular Detection Southeast Region



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UPDATES

- If you haven't yet – please register or submit an "intent to attend" for the in-person BTL training hosted by UGA on July 16-17.
 - There's an option for virtual attendees!
 - Please send one lead epidemiologist and one lead laboratorian

Sanibel 2.0



SANIBEL

BACTERIAL WGS ANALYSIS PIPELINE

Sanibel automatically runs species-specific typing modules (e.g., *Legionella*, *Klebsiella*, *Shigella*, *Salmonella*, *E. coli*, *Streptococcus*, *Pseudomonas*, *Acinetobacter*, *Vibrio*, *Neisseria*, and *Haemophilus*). Enhanced AMR and antigen analysis is provided for *N. meningitidis* and *H. influenzae*. PlasmidFinder is included for all samples.

- **Version 1:**
 - Core analysis
 - Basic processing steps
 - Standardized pipeline execution
- **Version 2:**
 - BMGAP2 integration
 - Improved modular workflow with better scalability
 - Updated dependencies
 - Enhanced reporting
 - Easier maintenance/troubleshooting



Biggest Problem!

“It works on my computer... but why doesn't it work on yours?”

- Dependency Conflicts
- Version Issues
- Hard Installs



What is Conda?

- Manages packages and environments
- Creates isolated workspaces
- Handles tools and dependencies
- Prevents software conflicts



Key Concepts

- Environment = Sandbox
- Package = tool/software



Why Conda is Important?

- Avoid conflicts
- Version Control
- Reproducibility
- Works with Nextflow





Conda > System Install

Feature	Conda	System Install
Isolated	✓	✗
Reproducible	✓	✗
Easier to Manage	✓	✗
Safer for Pipelines	✓	✗



Conda vs Mamba

- Conda = Standard Package Manager
- Mamba = Faster Conda Alternative



Installation Options

- Personal Computer → Miniconda / Mambaforge
- HPG / HPC → `module load conda`



Channels

- defaults → basic packages
- bioconda → bioinformatics tools
- conda-forge → community-maintained packages

```
conda config --add channels defaults  
conda config --add channels bioconda  
conda config --add channels conda-forge
```



Create Environment

```
conda create -n myenv  
conda activate myenv
```

Always create separate
environments



Install Packages

```
mamba install fastqc samtools
```

Packages install only inside
the active environment



View Environments

```
conda env list  
conda list
```

View environments and
installed packages



Remove Environment

```
conda remove --name my env --all
```

Delete and recreate
environments easily



Bioinformatics Example

```
mamba create -n shigatyper -c bioconda shigatyper
```

One environment per tool
or workflow



Conda + Nextflow

```
nextflow run pipeline -profile conda
```

Nextflow can
automatically manage
Conda environments





Conda vs Containers

Feature	Conda	Containers
Lightweight	✓	✗
Fully Reproducible	⚠	✓
Easy Setup	✓	⚠
Common in Pipelines	✓	✓

Modern workflows
increasingly use
containers



Best Practices

- Use one environment per workflow
- Use Mamba instead of Conda
- Avoid using base environment
- Keep and Export environments small



Export Environment

```
conda env export > env.yml  
conda env create -f env.yml
```

Share reproducible
environments with others



Common Mistakes

- Installing everything in base
- Mixing unrelated tools
- Missing channels
- Dependency conflicts
- Using outdated versions



Live Demo

1. Create Environment
2. Activate Environment
3. Install Packages
4. Run Tools
5. Export and Remove Environment



Summary

- Conda manages environments
- Mamba improves speed
- Environments improve reproducibility
- Essential for bioinformatics workflows





Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

bphl-sebioinformatics@flhealth.gov

Molly Mitchell, PhD

Bioinformatics Supervisor

Molly.Mitchell@flhealth.gov

Nikhil Reddy, MS

Bioinformatician

Nikhil.Yengala@flhealth.gov

Sam Bernhoft, MPH

Bioinformatician

Samantha.bernhoft@flhealth.gov